

2.11.2 CALIBRATION PROCEDURES

Before a PM10 monitoring program commences, it is essential to properly calibrate all sampling and laboratory equipment. Calibration is defined as the relationship between an instrumental output and the input of a known reference standard. Data that are traceable to common reference standards are more uniform in character and more readily comparable than data that are not traceable. Because PM10 concentration standards are not available for determining calibration relationships, individual components of the sampling method must be calibrated to ensure the integrity of reported data.

Like a TSP sampler, an HV PM10 sampler is essentially a device that pulls a sample of ambient air through a filter during a measured time period and collects particulate mass on the filter. Thus, to establish ambient PM10 concentrations, three independent determinations are made: air volume flow rate, sampling time, and particulate mass. The objective of this subsection is to provide technically sound flow-rate calibration procedures for both the MFC and VFC HV PM10 samplers.

Note: Calibration procedures for samplers that have been designated as reference methods will also be provided by the manufacturer in the operation or instruction manual. These manufacturer-provided calibration procedures are likely to be more specific and detailed than the more generic procedures presented herein. Also, minimum calibration requirements for the determination of particulate mass and sample time are presented in a table at the end of this subsection (Table 2.1).

Although it concerns TSP samplers rather than HV PM10 samplers, Reference 3 provides useful information concerning flow-rate-calibration procedures that is applicable to PM10 samplers. Reference 4 provides useful information concerning positive-displacement, standard volume meters (e.g., Roots^R Meters).

The following aspects of PM10 monitoring are discussed in this subsection:

- 2.1 A discussion of volumetric and mass flow measurements and their applicability in a PM10 monitoring program.
- 2.2 A certification procedure for an orifice flow-rate transfer standard.
- 2.3 Basic calibration procedure specific to an MFC HV PM10 sampler.
- 2.4 Basic calibration procedure specific to a VFC HV PM10 sampler.
- 2.5 Calibration frequency requirements.

Table 2.1, at the end of this subsection, presents minimum accuracy requirements for equipment that is used in PM10 sampling. It also references appropriate subsections in Section 2.2 of this volume.

Note: Consistency of temperature and barometric pressure units is required. It is recommended that all temperatures be expressed in kelvin ($K = ^\circ C + 273$). It is also recommended that all barometric pressures be expressed in either mm Hg or kPa

(but do not mix the two units). Take care to avoid calibrating a PM10 sampler using one set of units and then performing sample calculations using another set of units.

2.1 Discussion of Flow-Rate Measurement and General Aspects of PM10 Sampler Calibration

An HV PM10 sampler consists of two basic components: a specially designed inlet and a flow-rate controlling system. The particle size discrimination characteristics of both the impaction and cyclonic type inlets depend critically on maintaining certain air velocities within the inlet; a change in velocity will result in a change in the nominal particle size collected. For this reason, it is imperative that the flow rate through the inlet be maintained at a constant value that is as close as possible to the inlet's design flow rate. The design flow rate for a given sampler is specified in the sampler's instruction manual. The manual may also provide tolerance limits (or upper and lower limits) within which the sampler flow must be maintained. If the tolerance is not specified by the manufacturer, it should be assumed to be ± 10 percent. For example, if the design flow rate is specified as $1.13 \text{ m}^3/\text{min}$ with no tolerance given, the acceptable flow-rate range would be 1.02 to $1.24 \text{ m}^3/\text{min}$.

As indicated above, the true or actual flow rate through the sampler inlet must be known and controlled to ensure that only those particles nominally less than $10 \text{ }\mu\text{m}$ are being collected. A common source of error in a PM10 monitoring program is confusion of various air volume flow-rate measurement units. Although the sampler's operational flow rate must be monitored in terms of actual volume flow rate units (Q_a), sampler flow rates must be corrected to standard volume flow rate units (Q_{std}) at EPA standard conditions of temperature and pressure to calculate PM10 concentrations as required by EPA. Thus, both Q_a and Q_{std} flow rates are used for PM10 measurements. Before calibration procedures are initiated, the operating agency personnel should review the following flow-rate measurement designations:

- Q_a : Actual volumetric air flow rates, measured and expressed at existing conditions of temperature and pressure and denoted by Q_a (Q_{actual}). Typical units are L/min and m^3/min . Inlet design flow rates are always given in actual volumetric flow rate units.
- Q_{std} : Air flow rates that have been corrected to equivalent standard volume flow rates at EPA standard conditions of temperature and pressure ($25 \text{ }^\circ\text{C}$ or 298 K and 760 mm Hg or 101 kPa) and denoted by Q_{std} (Q_{standard}). Typical units are std. L/min and std. m^3/min . Standard volume flow-rate units are often used by engineers and scientists because they are equivalent to mass flow units. Standard volumes (derived from standard volume flow rates) are required for the calculation of PM10 mass concentration ($\mu\text{g}/\text{std. m}^3$) in reporting PM10 measurements.

These Q_a and Q_{std} measurement units must not be confused or interchanged. The flow rate units can be converted as follows, provided the existing temperature and pressure (or in some cases the average temperature and pressure over a sampling period) are known:

$$Q_{std} = Q_a(P_a/P_{std})(T_{std}/T_a) \quad (\text{Eq. 1})$$

$$\overline{Q_{std}} = \overline{Q_a}(P_{av}/P_{std})(T_{std}/T_{av}) \quad (\text{Eq. 1a})$$

$$Q_a = Q_{std}(P_{std}/P_a)(T_a/T_{std}) \quad (\text{Eq. 2})$$

where:

- Q_{std} = standard volume flow rate, std. m^3/min
- Q_a = actual volume flow rate, actual m^3/min
- P_a = ambient barometric pressure, mm Hg (or kPa)
- P_{std} = EPA standard barometric pressure, 760 mm Hg (or 101 kPa)
- T_{std} = EPA standard temperature, 298 K
- T_a = ambient temperature, K ($K = ^\circ\text{C} + 273$)
- $\overline{Q_{std}}$ = average standard volume flow rate for the sample period, std. m^3/min
- $\overline{Q_a}$ = average actual volume flow rate for the sample period, m^3/min
- P_{av} = average ambient barometric pressure during the sample period, mm Hg (or kPa)
- T_{av} = average ambient temperature during the sample period, K

Because it is critical to continually verify that the HV PM10 sampler flow rate is being maintained at close to the inlet design flow rate, the procedures in this subsection recommend that individual average temperature (T_{av}) and pressure (P_{av}) measurements or estimates be used for each 24-h sample period. However, if individual pressure and temperature values for each sample period cannot be obtained, seasonal average barometric pressure (P_s) and seasonal average temperature (T_s) for the site may be substituted to determine the sampler flow rate, provided that actual 24-h average temperatures at the site are not likely to deviate unreasonably from the seasonal average temperature used for the site.

The calibration procedures in this subsection provide for calibration of both the sampler and the flow rate transfer standard in actual volumetric flow-rate units (Q_a). This calibration mode allows convenient monitoring of the flow rate of the sampler against its design flow rate, is consistent with most sampler manufacturers' recommendations, and is in wide use. The actual volumetric flow rates of the sampler during sampling are then converted to the equivalent standard volumetric flow rates (Q_{std}), using Equation 1, for calculation of the PM10 concentration in $\mu\text{g}/\text{std. m}^3$. High-volume samplers for total suspended particulates (TSP) or lead may also be calibrated in this mode, using adaptations of these procedures, if desired for agency consistency. Other modes of calibration (such as calibration in terms of Q_{std}) are not precluded provided they produce accurate measurements of Q_a for comparing the sampler flow rate to its design flow rate, and of Q_{std} for calculating PM10 concentrations.

2.2 Certification of an Orifice Transfer Standard

The following certification procedure is applicable to an orifice transfer standard such as those that have been used previously in the calibration of TSP samplers. Two common types of orifice devices are available: one equipped with a

set of fixed resistance plates (e.g., a reference flow [ReF] device or a top-hat orifice), and the other with an externally variable resistance valve. The series of plates normally provided by the orifice manufacturer includes an 18-, 13-, 10-, 7-, and 5-hole plate. Unfortunately, the 5-hole plate provides too low a flow rate to be useful for HV PM10 calibration, and other plates may produce flow rates substantially outside the design flow-rate range of the commercially available HV PM10 inlets. Agencies may opt to fabricate or procure a different series of resistance plates that will provide more flow rates within the sampler's design flow-rate range or use the variable-resistance type orifice device.

2.2.1 Orifice Calibration Procedure -

1. Assemble the following equipment:

- Orifice transfer standard (i.e., top-hat orifice, variable orifice, or ReF device) to be calibrated.
- Water or oil manometer with a 0- to 400-mm (0- to 16-in.) range and minimum scale divisions of 2 mm (0.1 in.). This manometer should be permanently associated with the orifice transfer standard.
- Variable voltage transformer (or a set of resistance plates, if a variable voltage transformer is not available).
- Calibrated positive displacement, standard volume meter (such as a Roots^R Meter) traceable to the National Institute of Standards and Technology (NIST).

Note: As they are sold, standard volume meters may not be traceable to NIST. Traceability can be established directly through NIST or indirectly through the meter manufacturer's repair department. Periodic recertification is not normally required under clean service conditions unless the meter has been damaged and must be repaired. In general, damage will be indicated by a substantial (e.g., 50-percent) increase in the pressure drop across the meter. The meter's traceability certificate should contain a graph of the pressure drop as a function of flow rate. See Reference 4 for additional information concerning standard volume meters.

- High-volume air mover (e.g., a blower motor from a HV sampler).
- Accurate stopwatch.
- Mercury manometer, with a 0- to 200-mm (0- to 8-in.) range and minimum scale divisions of 2 mm (0.1 in.).
- Thermometer, capable of accurately measuring temperatures over the range of 0 to 50 °C (273 to 323 K) to the nearest ± 1 °C and referenced to an NIST or American Society for Testing and Materials (ASTM) thermometer within ± 2 °C at least annually.

- Barometer, capable of accurately measuring ambient barometric pressure over the range of 500 to 800 mm Hg (66 to 106 kPa) to the nearest mm Hg and referenced within ± 5 mm Hg of a barometer of known accuracy at least annually.
 - Orifice transfer standard certification worksheet such as shown in Figure 2.1 (blank forms appear in Subsection 12).
2. Record on the certification worksheet the standard volume meter's serial number; orifice transfer standard's type, model, and serial number; the person performing the certification; and the date.
 3. Observe the barometric pressure and record it as Pa.
 4. Read the ambient temperature in the vicinity of the standard volume meter and record it as T_a ($K = ^\circ C + 273$).
 5. Connect the orifice transfer standard to the inlet of the standard volume meter. Connect the mercury manometer to measure the pressure at the inlet of the standard volume meter. Connect the orifice (water or oil) manometer to the pressure tap on the orifice transfer standard. Connect a high-volume air mover to the outlet side of the standard volume meter. (See Figure 2.5 of Section 2.2 of this Handbook for an example of the calibration setup.) Make sure that all gaskets are present and are in good condition.
 6. Check that the standard volume meter table is level and adjust its legs if necessary.
 7. Check for leaks by temporarily clamping both manometer lines (to avoid fluid loss) and blocking the orifice with a large-diameter rubber stopper, wide duct tape, or other suitable means. Start the high-volume air mover and note any change in the standard volume meter's reading.

Caution: Avoid running the sampler for longer than 30 s at a time with the orifice blocked. This precaution will reduce the chance that the motor will be overheated due to the lack of cooling air. Such overheating can shorten the motor's lifetime. It can raise temperatures to the point of defeating the electrical insulation which could result in fire or electric shock to the user.

The reading should remain constant. If the reading changes, locate any leaks by listening for a whistling sound and/or retightening all connections, making sure that all gaskets are properly installed.

Note: The leak test procedure that is described above is one of several alternate procedures which may be used. Operating agencies may develop their own procedures.

8. After satisfactorily completing the leak check, turn off the high-volume air mover, unblock the orifice, and unclamp both manometer lines. Zero

ORIFICE TRANSFER STANDARD CERTIFICATION WORKSHEET						
Date: 7/22/88		Roots meter S/N: 7464318		Ta: 22.6 K 295.6		
Operator: G. PRATT		Orifice S/N: 2		Pa: 756.2 mm Hg		
Plate or Volts AC	Initial Volume	Final Volume	Δ Vol.	ΔTime (min)	ΔHg (mm)	ΔH ₂ O (in.)
1	5692	5695	3.0	3.41	18	3.0
2	5698	5701	3.0	2.75	25	4.5
3	5705	5708	3.0	2.34	32	6.0
4	5712	5715	3.0	2.00	42	8.0
5	5719	5722	3.0	1.88	48	9.0
DATA TABULATION						
Vstd	(x-axis) Qstd	(y-axis) [ΔH ₂ O (Pa/Ta)] ^{1/2} *	Va	(x-axis) Qa	(y-axis) [ΔH ₂ O (Ta/Pa)] ^{1/2}	
			2.929	0.859	1.083	
			2.901	1.055	1.327	
			2.873	1.228	1.532	
			2.833	1.417	1.769	
			2.810	1.495	1.876	
m =			m = 1.2390			
b =			b = 0.0172			
r =			r = 0.9999			
CALCULATIONS						
Vstd = Δ Vol [(Pa - ΔHg)/760] (298/Ta)*			Va = ΔVol [(Pa - ΔHg)/Pa]			
Qstd = Vstd/ΔTime*			Qa = Va/ΔTime			
y = mx + b			y = mx + b			
For subsequent flow rate calculations:						
Qstd = {[ΔH ₂ O (Pa/Ta)] ^{1/2} - b} {1/m}*			Qa = {[ΔH ₂ O (Ta/Pa)] ^{1/2} - b} {1/m}			

*NOTE: For PM10 monitoring, a calibration curve corrected to standard conditions is optional.

Figure 2.1. Example Orifice Transfer Standard Certification Worksheet.

the water and mercury manometers by sliding their scales so that their zero lines are even with the bottom of the meniscuses.

9. Turn on the high-volume air mover. Adjust the variable voltage transformer to achieve an appropriate flow rate (i.e., within the approximate range of 0.9 to 1.3 m³/min (32 to 46 ft³/min)). If necessary, use fixed resistance plates or the variable resistance valve to achieve the appropriate flow rate. The use of fixed resistance plates is discouraged (but not prohibited) because the leak check must be repeated each time that a plate is installed.
10. After setting a flow rate, allow the system to run for at least 1 min to attain a constant motor speed. Observe the standard volume meter dial reading and simultaneously start the stopwatch. Error in reading the meter dial can be minimized by starting and stopping the stopwatch on whole number dial readings (e.g., 4091.00).
11. Record the initial volume that the meter dial was indicating when the stopwatch was started. Maintain this constant flow rate until at least 3 m³ of air have passed through the standard volume meter. Record the standard volume meter's inlet pressure manometer reading as ΔHg , and the orifice manometer reading as ΔH_2O . Be sure to indicate the correct units of measurement. If ΔH_2O changes significantly during the run, abort the run and start again.
12. When at least 3 m³ of air have passed through the system, note the standard volume meter reading and simultaneously stop the stopwatch. Record the final volume that the meter dial was indicating when the stopwatch was stopped. Record the elapsed time ($\Delta Time$) indicated on the stopwatch.
13. Calculate the volume measured by the standard volume meter ($\Delta Vol.$) using Equation 3 and record.

$$\Delta Vol. = \text{Final Volume} - \text{Initial Volume} \quad (\text{Eq. 3})$$

14. Correct this volume to ambient atmospheric pressure.

$$V_a = \Delta Vol. (P_a - \Delta Hg) / P_a \quad (\text{Eq. 4})$$

where:

- V_a = actual volume at ambient barometric pressure, m³
- $\Delta Vol.$ = actual volume measured by the standard volume meter, m³
- P_a = ambient barometric pressure during calibration, mm Hg (or kPa)
- ΔHg = differential pressure at inlet to volume meter, mm Hg (or kPa).

15. Calculate the actual volumetric flow rate (m^3/min).

$$Q_a = V_a / \Delta \text{Time} \quad (\text{Eq. 5})$$

where:

Q_a = actual volumetric flow rate through the orifice, m^3/min
 V_a = actual volume at ambient barometric pressure, m^3
 ΔTime = elapsed time, min.

16. Repeat Steps 9 through 15 for at least four additional flow rates within the approximate range of 0.9 to 1.3 m^3/min (32 to 46 ft^3/min). At least five evenly distributed different flow rates are required, and at least three flow rates must be in the specified inlet flow-rate interval [1.02 to 1.24 m^3/min (36 to 44 ft^3/min)]. Better calibration precision may be obtained by running additional flow rates or repeating the flow rates.
17. For each flow, compute $[(\Delta H_2O)(T_a/P_a)]^{1/2}$, and plot these values against the corresponding values of Q_a . Draw the orifice transfer standard's certification curve. For the model $[(\Delta H_2O)(T_a/P_a)]^{1/2} = m(Q_a) + b$, calculate the linear least squares regression's slope (m), intercept (b), and correlation coefficient (r) of the certification relationship. Plot the regression line on the same graph as the calibration data. A certification graph should be readable to 0.02 m^3/min .
18. If any calibration point does not fall within ± 2 percent of the line, rerun that point, recalculate, and replot.
19. For subsequent use of the orifice transfer standard, calculate Q_a from the calibration relationship as:

$$Q_a(\text{orifice}) = \{[(\Delta H_2O)(T_a/P_a)]^{1/2} - b\} \{1/m\} \quad (\text{Eq. 6})$$

where:

$Q_a(\text{orifice})$ = actual volumetric flow rate as indicated by the orifice transfer standard, m^3/min
 ΔH_2O = pressure drop across the orifice, mm (or in.) H_2O
 T_a = ambient temperature during use, K ($K = ^\circ\text{C} + 273$)
 P_a = ambient barometric pressure during use, mm Hg (or kPa)
 b = intercept of the orifice calibration relationship
 m = slope of the orifice calibration relationship.

2.2.2 Orifice Transfer Standard Calibration Frequency -

Upon receipt and at 1-yr intervals, the calibration of the orifice transfer standard should be certified with a standard volume meter (such as a Roots^R Meter) traceable to NIST. An orifice transfer standard should be visually inspected for signs of damage before each use, and should be recalibrated if the inspection reveals any nicks or dents.

2.3 Basic Calibration Procedure for a Mass-Flow-Controlled (MFC) Sampler Using an Orifice Transfer Standard

The MFC sampler calibration procedure presented in this subsection relates known flow rates to the pressure in the exit orifice plenum. The known flow rates are determined by an orifice transfer standard that has been certified according to the procedure presented in Section 2.2.1. The exit orifice plenum is the area within the motor housing (below the motor unit) that contains the air flow just before it is exhausted to the atmosphere through the exit orifice. It is recommended that this exit orifice plenum pressure be measured with a 25-cm (10-in.) water or oil manometer. It is further recommended that each sampler should have its own dedicated manometer, which can be conveniently mounted to the side of the sampler housing. Other types of pressure measurement devices may be used provided they have comparable accuracy. However, the 4-in. continuous pressure (flow) recorders of the type often supplied with HV PM10 samplers are generally not sufficiently accurate and are not recommended for quantitative sampler pressure or flow measurements. These flow recorders should be used only for nonquantitative determination that the flow was approximately constant and uninterrupted over the sample period. The flow recorder may be connected in parallel with the manometer or other pressure measuring device, using a tee or "Y" tubing connection.

Note: Because flow recorders are still in wide use for quantitative flow measurements, the calibration procedure includes specific instructions for quantitatively calibrating a flow recorder. These flow recorder instructions are enclosed in brackets [] and should be used only when a manometer or other pressure measurement device cannot be used.

For this MFC calibration procedure, the following conditions are assumed:

- The HV PM10 sampler is equipped with a mass flow controller to control its sample flow rate.
- The sampler flow rate is measured by measuring the exit orifice plenum pressure, using a water or oil manometer [or, if necessary, a continuous-flow recording device using square-root-scale chart paper].
- The sampler inlet is designed to operate at an actual volumetric flow rate of 1.13 m³/min, and the acceptable flow-rate range is ± 10 percent of this value.
- The transfer standard for the flow-rate calibration is an orifice device equipped with either a series of resistance plates or an integral variable-resistance valve. The pressure drop across the orifice is measured by an associated water or oil manometer.
- The sampler will be calibrated in actual volumetric flow-rate units (Qa), and the orifice transfer standard is also calibrated in Qa, as specified in Subsection 2.2.

2.3.1 Calibration Equipment -

1. Orifice transfer standard with calibration traceable to NIST (see Subsection 2.2).

Note: The predrilled resistance plates that are supplied with the orifice transfer standard may have to be modified for the calibration of PM10 samplers. The holes in the plates may have to be enlarged or additional holes may have to be drilled to obtain flow rates within the acceptable range. Alternately, new resistance plates could be purchased if they are available.

2. An associated water or oil manometer, with a 0- to 400-mm (0- to 16-in.) range and a minimum scale division of 2 mm (0.1 in.).
3. A water or oil manometer, with a 0- to 200-mm (0- to 8-in.) range and a minimum scale division of 2 mm (0.1 in.) for measurement of the sampler exit orifice plenum pressure. This manometer should be associated with the sampler.

Note: Manometers used for field calibration may be subject to damage or malfunction and should thus be checked frequently.

4. Thermometer, capable of accurately measuring temperature over the range of 0 to 50 °C (273 to 323 K) to the nearest ± 1 °C and referenced to an NIST or ASTM thermometer within ± 2 °C at least annually.
5. A portable aneroid barometer (e.g., a climber's or engineer's altimeter) capable of accurately measuring ambient barometric pressure over the range of 500 to 800 mm Hg (66 to 106 kPa) to the nearest mm Hg and referenced within ± 5 mm Hg of a barometer of known accuracy at least annually.
6. Miscellaneous handtools, calibration data sheets or station log book, and 51-mm (2-in.) duct tape.

2.3.2 Multipoint Flow-Rate Calibration Procedure - MFC Sampler -

The procedure presented here is basic and generic, given the assumptions listed in Subsection 2.3. There may be more detailed calibration procedures, variations, or alternative calibration procedures presented in the manufacturer's instruction manual. It is recommended that the manual be reviewed carefully and that the various calibration variations or alternative procedures be evaluated. In-house equipment and personnel, procedural simplicity and uniformity, and subsequent data applications should be considered in establishing the specific, detailed calibration procedure to be implemented.

Caution: Do not attempt to calibrate the MFC sampler under windy conditions. Short-term wind velocity fluctuations will produce variable pressure readings by the orifice transfer standard's manometer. The calibration will be less precise because of the pressure variations.

1. Set up the calibration system as recommended by the manufacturer. Figure 2.2 illustrates the calibration configuration of a typical MFC sampler. MFC samplers are calibrated without a filter or filter cassette installed.
2. Disconnect the motor from the flow controller and plug it directly into a stable line voltage source (i.e., the sampler's on-off timer, if so equipped, or other source of the line voltage).
3. Install the orifice transfer standard and its adapter faceplate on the sampler. Check all gaskets and replace any questionable ones.

Caution: Tighten the faceplate nuts evenly on alternate corners to properly align and seat the gaskets. The nuts should be only hand-tightened because too much compression can damage the sealing gasket.

4. Select the first calibration flow rate and install the appropriate resistance plate or adjust the variable orifice valve. At least four flow rates are required to define the calibration relationship.

At least three flow rates should be within (or nearly within) the acceptable flow-rate range (i.e., 1.02 to 1.24 m³/min) for the sampler inlet. For resistance plate orifices, make sure that the orifice and resistance plate gaskets are in place and the orifice is not cross-threaded on the faceplate.

5. Leak Test: Block the orifice with a large-diameter rubber stopper, wide duct tape, or other suitable means. Seal the pressure port with a rubber cap or similar device. Turn on the sampler.

Caution: Avoid running the sampler for longer than 30 s at a time with the orifice blocked. This precaution will reduce the chance that the motor will be overheated due to the lack of cooling air. Such overheating can shorten the motor's lifetime. It can raise temperatures to the point of defeating the electrical insulation, which could result in fire or electric shock to the user.

Gently rock the orifice transfer standard and listen for a whistling sound that would indicate a leak in the system. A leak-free system will not produce an upscale response in the sampler's exit orifice manometer or flow recorder. Leaks are usually caused either by a damaged or missing gasket between the orifice transfer standard and the faceplate or by cross-threading of the orifice transfer standard on the faceplate. All leaks must be eliminated before proceeding with the calibration. When the system is determined to be leak-free, turn off the sampler and unblock the orifice.

Note: The leak test procedure that is described above is one of several alternate procedures which may be used. Operating agencies may develop their own procedures.

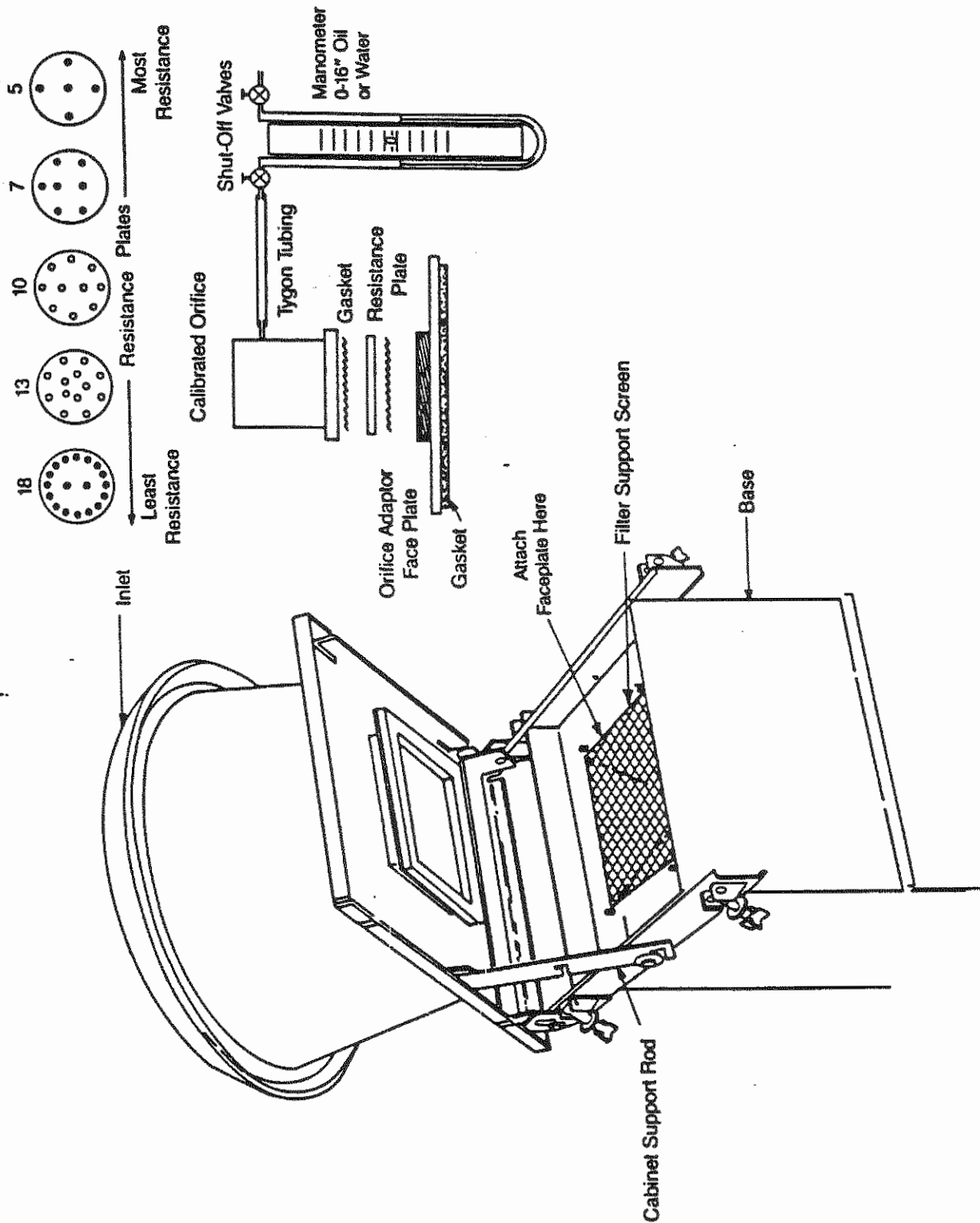


Figure 2.2. MFC sampler calibration equipment.

6. Inspect the connecting tubing of both manometers for crimps or cracks. Open the manometer valves (if present) and blow gently through the tubing, watching for the free flow of the fluid.

Adjust the manometers' sliding scales so that their zero lines are at the bottom of the menisci. Connect the orifice transfer standard manometer to the orifice transfer standard. Connect the sampler's exit orifice manometer [and the continuous-flow recorder, if used] to the exit orifice plenum port. Ensure that one side of each manometer is open to atmospheric pressure. Make sure that the tubing fits snugly on the pressure ports and on the manometer.

7. [If a continuous flow recorder is to be used quantitatively in lieu of a manometer, record the site location, sampler S/N, date, and the operator's initials on the blank side of a clean recorder chart. Make sure the chart has a square-root scale. Open the front door of the sampler and install the clean recorder chart.]
8. Read and record the following parameters on the HV PM10 data sheet. Figure 2.3 presents an example calibration data sheet for the MFC sampler (blank forms appear in Subsection 12).

- Date, location, and operator's signature.
- Sampler S/N and model.
- Ambient barometric pressure (Pa), mm Hg or kPa.
- Ambient temperature (T_a), K ($K = ^\circ C + 273$).
- Orifice S/N and calibration relationship.

Note: Consistency of temperature and barometric pressure units is required. It is recommended that all temperatures be expressed in kelvin ($K = ^\circ C + 273$). It is also recommended that all barometric pressures be expressed in either mm Hg or kPa (but do not mix the two units). Take care to avoid calibrating a PM10 sampler using one set of units and then performing sample calculations using another set of units.

Note: Ideally, the temperature of the air in the exit orifice plenum should be measured because it will be somewhat higher than ambient temperature. However, an adequate approximation of this temperature may be obtained by adding 30 K to the ambient temperature. This addition is incorporated in the calculations given in Subsection 2.3.3.

9. Turn on the sampler and allow it to warm up to operating temperature (3 to 5 min). Then, read and record the orifice transfer standard's manometer deflection, ΔH_2O (in. H_2O), and the corresponding sampler's manometer deflection, ΔP_{ex} [or flow recorder chart reading, I].

Note: The sampler inlet may be partially lowered over the orifice transfer standard to act as a draft shield (if a shield is not otherwise provided). Use a block to provide at least 2 in. of clearance at the bottom for air flow and for the manometer tubing.

MFC SAMPLER CALIBRATION DATA SHEET

Station Location MILFORD, NC Date 10/15/88 Time 0900
Sampler Model GMW S/N 1297 Operator D. PALMER
Pa 762 mm Hg, Ta 19.4 °C 292.4 K, Unusual conditions: NONE
Ps 743 mm Hg, Ts 8.7 °C 281.7 K, (*seasonal average Ta and Pa)
Orifice S/N 2 Orifice Calibration Date 7/22/88
Orifice calibration relationship: $m = 1.2390$ $b = 0.0172$ $r = 0.9999$

Plate Number	Total ΔH_2O (in.)	X-Axis = Q_a (orifice) flow rate ^a (m^3/min)	Sampler ΔP_{ex} (in. H_2O) [or I for flow recorders]	Y-Axis = Sampler ΔP_{ext} ^b [or It for flow recorders] ^c
1	8.34	1.430	5.8	1.57
2	7.55	1.360	5.4	1.51
3	6.59	1.270	4.7	1.41
4	5.80	1.190	4.4	1.36
5	4.61	1.060	3.6	1.23

$$^a Q_a = \{[(\Delta H_2O) (T_a/P_a)]^{1/2} - b\} \{1/m\}$$

$$^b \Delta P_{ext} = [\Delta P_{ex}(T_a + 30)/P_a]^{1/2}$$

$$^c It = [I] [(T_a + 30)/P_a]^{1/2} \text{ if a flow recorder is used}$$

Sampler Calibration Relationship (Q_a on x-axis; ΔP_{ext} or It on y-axis):

$$\Delta P_{ext} = m[Q_a (\text{Orifice})] + b \text{ or } It = m[Q_a (\text{Orifice})] + b$$

$$m = 0.9137 \quad b = 0.2629 \quad r = 0.9979$$

For subsequent calculation of sampler flow rate:

$$Q_a = \{[\text{mean } \Delta P_{ex} (T_{av} + 30)/P_{av}]^{1/2} - b\} \{1/m\}$$

$$\text{or } Q_a = \{\text{mean } I[(T_{av} + 30)/P_{av}]^{1/2} - b\} \{1/m\}$$

$$\text{Set point flow rate (SFR)} \quad 1.14$$

$$\text{SFR} = 1.13 (P_s/P_a) (T_a/T_s)$$

$$\text{Sampler set point (SSP)} \quad 4.02$$

$$\text{SSP} = [P_a/(T_a + 30)] [m(\text{SFR}) + b]^2$$

or $\text{SSP} = [P_a/(T_a + 30)]^{1/2} [m(\text{SFR}) + b]$ for flow recorders

Figure 2.3. Example MFC sampler calibration data sheet.

10. Install the other resistance plates or adjust the variable orifice value to obtain each of the other calibration flow rates and repeat Step 9 for each. At least four calibration flow rates are required, with at least three in the acceptable flow-rate range (i.e., 1.02 to 1.24 m³/min).
11. Plot the calibration data on a sheet of graph paper as specified in Step 4 of the next subsection.

Note: The data should be plotted in the field as the calibration is occurring, rather than afterwards back at the laboratory.

Repeat Step 9 for any data that are questionable on the plot. Running additional calibration points at differing flow rates or repeating the calibration points at the same flow rates is encouraged to improve the precision of the calibration.

12. Turn off the sampler and remove the orifice transfer standard.
13. Reconnect the sampler motor to the flow controller.
14. Perform the calibration calculations presented in the following subsection. The data generated will be used to set the mass flow controller (Subsection 2.3.4) to a value that will result in optimal volumetric flow based on the seasonal average temperature and barometric pressure at the monitoring site.

2.3.3 Calibration Calculations -

- Gather together all the calibration data, including the orifice calibration information, the sampler calibration data sheet, [and, if used, the flow recorder chart, which should graphically display the various calibration flow rates].

Note: These calculations should be done at the time of the calibration, rather than later. This approach will allow additional calibration points to be taken if questions arise about the data that have already been obtained.

1. Verify that the orifice transfer standard calibration relationship is current and traceable to an acceptable primary standard.
2. Calculate and record Q_a for each calibration point from the orifice calibration information and Equation 6:

$$Q_a(\text{orifice}) = \{[\Delta H_2O(T_a/P_a)]^{1/2} - b\} \{1/m\} \quad (\text{Eq. 6})$$

where:

Q_a(orifice) = actual volumetric flow rate as indicated by the transfer standard orifice, m³/min

ΔH_2O = pressure drop across the orifice, mm (or in.) H_2O
 T_a = ambient temperature during use, K ($K = ^\circ C + 273$)
 P_a = ambient barometric pressure during use, mm Hg (or kPa)
 b = intercept of the orifice calibration relationship
 m = slope of the orifice calibration relationship.

3. Calculate and record the quantity ΔP_{ext} for each calibration point as:

$$\Delta P_{ext} = [\Delta P_{ex}(T_a+30)/P_a]^{1/2} \quad (\text{Eq. 7})$$

where:

ΔP_{ext} = transformed manometer reading
 ΔP_{ex} = sampler manometer reading, mm (or in.) H_2O
 T_a = ambient temperature, K ($K = ^\circ C + 273$)
 P_a = ambient barometric pressure, mm Hg (or kPa).

[If a continuous-flow recorder is used quantitatively, calculate and record the quantity I_t as follows:

$$I_t = I[(T_a+30)/P_a]^{1/2} \quad (\text{Eq. 7a})$$

where:

I_t = transformed flow recorder chart reading
 I = flow recorder chart reading, arbitrary units on square root scale.]

Note: If recorder charts with linear scales are used, substitute $(I)^{1/2}$ for I in Equation 7a.

4. On a sheet of graph paper, plot the calculated $Q_a(\text{orifice})$ flow rates on the x-axis vs. the transformed sampler manometer response, ΔP_{ext} [or the transformed flow recorder reading, I_t] on the y-axis.

Because the determination of the sampler's average operational flow rate (Q_a) during a sample period depends on the ambient average temperature and pressure, use of a graphic plot of the calibration relationship is not recommended for subsequent data reduction. This plot is used only to visually assess the calibration points to see if any should be rerun.

Plot the regression line on the same graph paper as the calibration data. For the regression model $y = mx + b$, let $y = \Delta P_{ext}$ and $x = Q_a(\text{orifice})$ so that the model is given by:

$$\Delta P_{ext} = m[Q_a(\text{orifice})] + b \quad (\text{Eq. 8})$$

[For the flow recorder, the model is:

$$I_t = m[Q_a(\text{orifice})] + b \quad (\text{Eq. 8a})]$$

Using a programmable calculator or a calculation data form, determine the linear regression slope (m), intercept (b), and correlation coefficient (r) and record them on the data sheet. A five-point calibration should yield a regression equation with a correlation coefficient of $r > 0.990$, with no point deviating more than $\pm 0.04 \text{ m}^3/\text{min}$ from the value predicted by the regression equation. Plot the regression line on the same graph paper that has the individual calibration points.

5. For subsequent sample periods, the sampler's average actual operational flow rate, $\overline{Q_a}$, is calculated from the calibration slope and intercept using Equation 9:

$$\overline{Q_a} = \{[\overline{\Delta P_{ex}}(T_{av}+30)/P_{av}]^{1/2} - b\} \{1/m\} \quad (\text{Eq. 9})$$

where:

$\overline{Q_a}$ = the sampler's average actual flow rate, m^3/min
 $\overline{\Delta P_{ex}}$ = average of initial and final sampler manometer readings, $(\Delta P_{ex_i} + \Delta P_{ex_f})/2$, mm (or in.) H_2O
 T_{av} = average ambient temperature for the sample period, K ($K = ^\circ\text{C} + 273$)
 P_{av} = average ambient pressure for the sample period, mm Hg (or kPa)
 b = intercept of the sampler calibration relationship
 m = slope of the sampler calibration relationship.

[For the flow recorder,

$$\overline{Q_a} = \{\overline{I}[(T_{av}+30)/P_{av}]^{1/2} - b\} \{1/m\} \quad (\text{Eq. 9a})$$

where:

$\overline{I}$ = average flow recorder reading for the sample period.]

Note: If recorder charts with linear scales are used, substitute $(I)^{1/2}$ for (I) in Equation 9a.

2.3.4 Mass Flow Controller Adjustment Procedure -

Because the controlled flow rate of an MFC sampler is adjustable, it must be set to the proper flow rate for the inlet. The constant mass flow maintained by the MFC causes the actual volumetric flow rate through the inlet to fluctuate as the ambient temperature and barometric pressure change at the monitoring site. Normally, the range of these fluctuations is within the allowable tolerance limits for the inlet. However, the flow-rate set point of the mass flow controller must be correctly adjusted so that the deviations are "centered" with respect to the seasonal average temperature and barometric pressure at the site, not the temperature and pressure prevailing at the time of setting. The correct volumetric set-point flow rate (SFR) at T_a and P_a has the same mass flow rate as the inlet design volumetric flow rate has at T_s and P_s .

Note: The correct SFR thus may be different from day to day and may be somewhat higher or lower than the inlet design flow rate on any particular day.

Set the mass flow controller as follows:

1. Determine the seasonal average temperature (T_s) and seasonal average pressure (P_s) at the site and record them on the calibration data sheet. (Determination of the number of "seasons," i.e., the number of different seasonal average temperatures needed for the year, is left to the discretion of the local agency.)

2. Calculate SFR and record on the calibration data sheet:

$$\text{SFR} = (1.13) (P_s/P_a)(T_a/T_s) \quad (\text{Eq. 10})$$

where:

SFR = set-point actual volumetric flow rate for adjustment of the mass flow controller, based on seasonal average temperature and average pressure at site, m^3/min

1.13 = inlet design flow rate (as specified by the manufacturer), m^3/min

P_s , P_a = seasonal average and current ambient barometric pressure at the site, respectively, mm Hg (or kPa)

T_s , T_a = seasonal average and current ambient temperature, respectively, K ($K = ^\circ\text{C} + 273$).

3. Calculate and record on the sampler's calibration data sheet the sampler set-point manometer reading [or flow recorder reading] that corresponds to the SFR calculated in Step 2.

$$\text{SSP} = [P_a/(T_a + 30)][m(\text{SFR}) + b]^2 \quad (\text{Eq. 11})$$

where:

SSP = sampler set-point manometer reading, mm (or in.) H_2O

P_a = ambient barometric pressure, mm Hg (or kPa)

T_a = ambient temperature, K ($K = ^\circ\text{C} + 273$)

m = slope of the sampler's calibration relationship

SFR = set-point flow rate from Equation 10, m^3/min

b = intercept of the sampler's calibration relationship.

[For the flow recorder,

$$\text{SSP} = [m(\text{SFR}) + b] [P_a/(T_a+30)]^{1/2} \quad (\text{Eq. 11a})$$

4. Visually check to make sure the motor is connected to the mass flow controller and the manometer is properly connected.
5. Install a clean filter (in a filter cassette) in the sampler according to the manufacturer's instructions.

[If the continuous flow recorder is used quantitatively, install a clean chart and verify that the recorder is zeroed (i.e., the pen rests on the innermost circle of the chart).]

6. Turn on the sampler and allow it to warm up to operating temperature (3 to 5 min).
7. Following the manufacturer's instructions, adjust the mass flow controller until the manometer reading [or flow recorder response] indicates the sampler set point (SSP) as calculated in Step 3.
8. Verify that the flow controller will maintain this flow rate for at least 10 min. Turn off the sampler.
9. The sampler can now be prepared for the next sample run day.

2.4 Basic Calibration Procedure for a Volumetric-Flow-Controlled (VFC) Sampler Using an Orifice Device

The VFC sampler calibration procedure presented in this subsection relates known flow rates (Q_a , as determined by an orifice transfer standard) to the ratio of the stagnation pressure to the ambient barometric pressure (P_1/P_a). The stagnation pressure (P_1) is the air pressure inside the sampler in the area just under the filter. VFC samplers have a stagnation pressure tap or port through which the stagnation pressure can be measured. A VFC sampler may also have an exit orifice below the motor similar to those in MFC samplers. In this case, the sampler flow rate could be measured and calibrated using the exit orifice plenum pressure as described in Subsection 2.3. However, the use of the stagnation pressure generally provides a more accurate indication of sampler flow rate. Additionally, a continuous-flow recorder may be connected to the exit orifice pressure tap for nonquantitative determination that the flow rate was constant and uninterrupted over the sample period.

It is recommended that the stagnation pressure be measured with a 0- to 1000-mm (0- to 36-in.) oil or water manometer (to avoid the hazards of a mercury manometer). It is further recommended that each sampler have its own dedicated manometer, which can be conveniently mounted to the side of the sampler housing. Other types of pressure measurement instruments may be used provided they have comparable accuracy. However, the 4-in. continuous pressure (i.e., flow) recorders often supplied with HV PM10 samplers are generally not sufficiently accurate and are not recommended for quantitative sampler pressure or flow rate measurements.

The VFC sampler's flow control system is a choked-flow venturi. It must be precisely sized for a given average annual temperature and pressure because no means is provided for the user to adjust the operational flow rate. Therefore, the purchasing agency should notify the manufacturer of the operational location of the sampler; differences in temperature and pressure between the shipping address and the monitoring site may result in an incorrect operational flow rate. As with the MFC sampler, it will be necessary to determine or estimate both the ambient

temperature and barometric pressure readings during the sample period for the subsequent calculation of total sample volume in standard volume units.

For this VFC calibration procedure, the following conditions are assumed:

- The VFC sampler uses a choked-flow venturi to control the actual volumetric flow rate.
- The sampler flow rate is measured by measuring the stagnation pressure ratio, and the sampler is not equipped with a continuous flow recorder.
- The sampler inlet is designed to operate at a constant actual volumetric flow rate of 1.13 m³/min, and the acceptable flow-rate range is ± 10 percent of this value.
- The transfer standard for the flow-rate calibration is an orifice device equipped with either a series of resistance plates or an integral variable-resistance valve. The pressure drop across the orifice is measured by an associated water or oil manometer.
- The sampler will be calibrated in actual volumetric flow-rate units (Qa), and the orifice transfer standard is also calibrated in Qa, as specified in Subsection 2.2.

2.4.1 Calibration Equipment -

1. Orifice transfer standard with proper calibration traceable to NIST (see Subsection 2.2).

Note: The predrilled resistance plates that are supplied with the orifice transfer standard may have to be modified for the calibration of PM10 samplers. The holes in the plates may have to be enlarged or additional holes may have to be drilled to obtain flow rates within the acceptable range. Alternatively, new resistance plates could be purchased if they are available.

2. An associated water or oil manometer, with a 0- to 400-mm (0- to 16-in.) range and minimum scale divisions of 2 mm (0.1 in.) for measurement of transfer standard pressure.
3. An oil or water manometer, with a 0- to 1000-mm (0- to 36-in.) range and minimum scale divisions of 2 mm (0.1 in.) or other pressure measurement device for measurement of the sampler stagnation pressure. Ideally, this manometer (or other pressure instrument) should be associated with the sampler.

Note: Manometers used for field calibration may be subject to damage or malfunction and should thus be checked frequently.

4. Thermometer, capable of accurately measuring temperature over the range of 0 to 50 °C (273 to 323 K) to the nearest ± 1 °C and referenced to an NIST or ASTM thermometer within ± 2 °C at least annually.

5. A portable, aneroid barometer (e.g., a climber's or engineer's altimeter), capable of accurately measuring ambient barometric pressure over the range of 500 to 800 mm Hg (66 to 106 kPa) to the nearest mm Hg and referenced within ± 5 mm Hg to a barometer of known accuracy at least annually.
6. Calibration data sheets or the station log book and 51-mm (2-in.)-wide duct tape.
7. A clean filter.

2.4.2 Multipoint Flow-Rate Calibration Procedure - VFC Sampler -

The procedure presented here is basic and intended to be generic, given the assumptions listed in Subsection 2.4. There may be more detailed calibration procedures, variations, or alternative calibration procedures presented in the manufacturer's instruction manual. It is recommended that the manual be reviewed carefully and that the various calibration variations or alternative procedures be evaluated. In-house equipment and personnel, procedural simplicity and uniformity, and subsequent data applications should be considered in establishing the specific, detailed calibration procedure to be implemented.

Note: The calibration of some VFC samplers may be affected by changes in line voltage, particularly if the line voltage is below normal (normal is about 115 VAC). For this reason, VFC samplers should always be calibrated at the monitoring site. Further, if the line voltage at the site is low and likely to fluctuate significantly, a line voltage booster or regulator may be advisable. Also, be sure that replacement blower motors are of the correct type. Avoid the use of lower-amperage blower motors that are intended for use in HV TSP samplers.

Caution: Do not attempt to calibrate the VFC sampler under windy conditions. Short-term velocity fluctuations will produce variable pressure readings by the orifice transfer standard's manometer. The calibration will be less precise because of the pressure variations.

1. Set up the calibration system as recommended by the manufacturer. Figure 2.4 illustrates the calibration configuration of a typical VFC sampler. The VFC sampler manufacturer may specify that the sampler be calibrated with a filter installed, which generally precludes calibration flow rates higher than normal operating flow rate. Additional calibration flow rates obtained without a filter may be appropriate, as discussed in Step 8.
2. Install the orifice transfer standard and its adapter faceplate on the sampler: First inspect all gaskets and seals, and replace any doubtful ones.

Caution: Tighten the faceplate nuts evenly on alternate corners to properly align and uniformly seat the gaskets. The nuts should be hand-tightened only; too much compression can damage the sealing gasket.

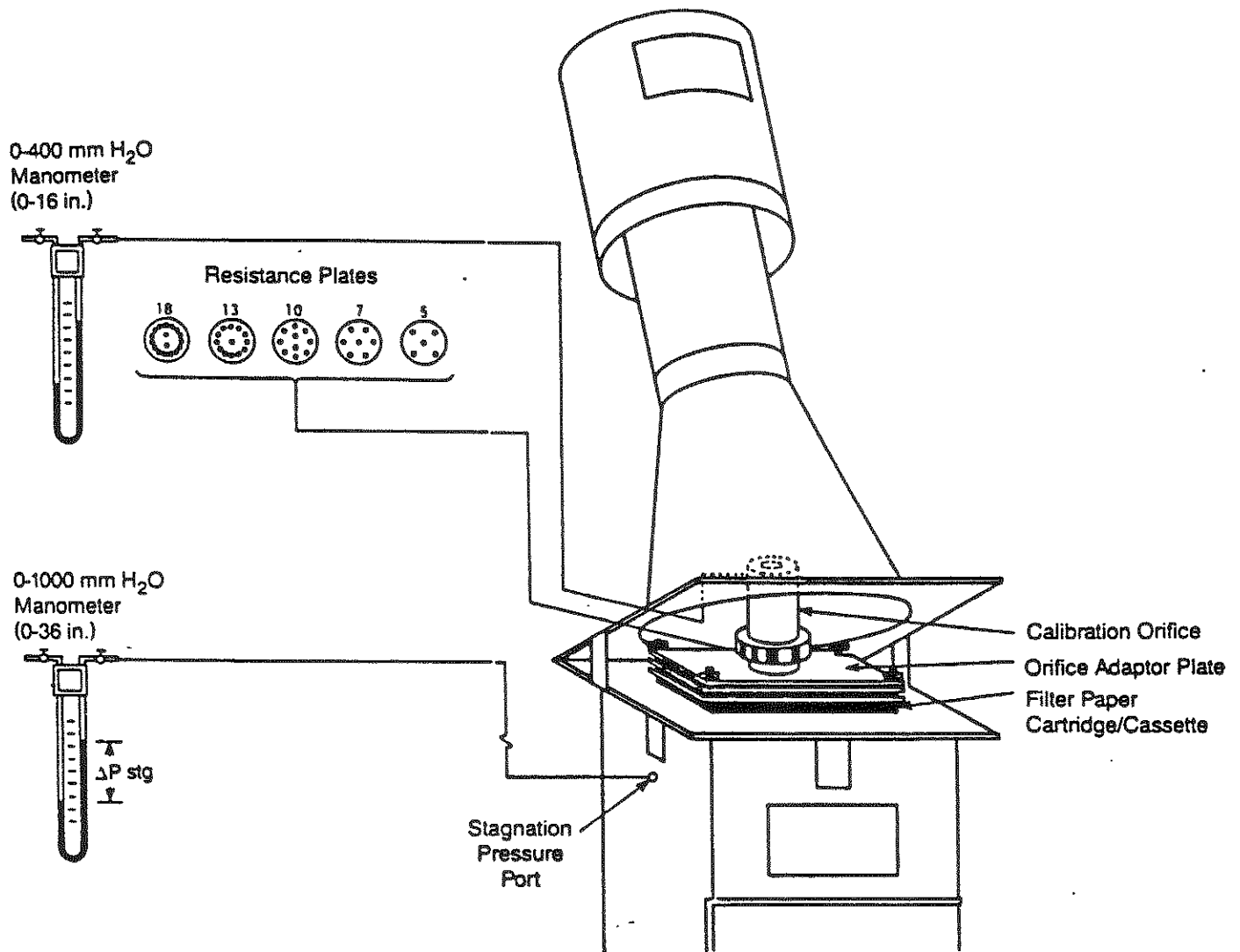


Figure 2.4. Calibration assembly of the VFC sampler.

3. Select a calibration flow rate and install the appropriate resistance plate (or no plate) or adjust the variable resistance valve. At least four flow rates are required to define the calibration relationship. At least three flow rates should be within the acceptable flow-rate range (i.e., 1.02 to 1.24 m³/min) for the sampler inlet. For resistance plate orifices, make sure the orifice and resistance plate gaskets are in place and the orifice is not cross-threaded on the faceplate.
4. Leak Test: Block the orifice with a large-diameter rubber stopper, wide duct tape, or other suitable means. Seal both orifice and stagnation pressure ports with rubber caps or similar devices. Turn on the sampler.

Caution: Avoid running the sampler for longer than 30 s at a time with the orifice blocked. This precaution will reduce the chance that the motor will be overheated due to the lack of cooling air. Such overheating can shorten the motor's lifetime. It can raise temperatures to the point of defeating the electrical insulation, which could result in fire or electric shock to the user.

Gently rock the orifice transfer standard and listen for a whistling sound that would indicate a leak in the system. Leaks are usually caused either by a damaged or missing gasket between the orifice transfer standard and the faceplate or by crossthreading of the orifice transfer standard on the faceplate. All leaks must be eliminated before proceeding with the calibration. When the system is determined to be leak-free, turn off the sampler and unblock the orifice.

Note: The leak test procedure that is described above is one of many alternate procedures which may be used. Operating agencies may develop their own procedures.

5. Inspect the connecting tubing of the manometers for crimps or cracks. Open the manometer valves (if present) and blow gently through the tubing, watching for the free flow of the fluid.

Adjust the manometers' sliding scales so that their zero lines are at the bottom of the menisci. Connect the transfer standard manometer to the transfer standard and the sampler stagnation pressure manometer (or other pressure instrument) to the stagnation pressure port. Ensure that one side of each manometer is open to atmospheric pressure. Make sure the tubing fits snugly on the pressure ports and on the manometers.

6. Read and record the following parameters on the VFC Sampler Data Sheet. Figure 2.5 presents an example calibration data sheet for the VFC sampler (blank forms appear in Subsection 12).
 - Date, location, and operator's signature.
 - Sampler S/N and model.

VFC SAMPLER CALIBRATION DATA SHEET

Station Location GLENDORA, CA Date 5/1/89 Time 2:30 AM
 Sampler Model WEDDING S/N 0340870172P Operator T. WISBETH
 Pa 755 mm Hg, Ta 22.2 °C 295.2 K, Unusual Conditions NONE
 Orifice S/N 2 Orifice Calibration Date 7/22/88
 Orifice Calibration Relationship: $m =$ 1.2390 $b =$ 0.0172 $r =$ 0.9999

Plate No.	ΔH_2O (in.)	ΔP_{stg} (mm Hg) ^a	$P1 = Pa - \Delta P_{stg}$ (mm Hg)	$P1/Pa$ (mm Hg)	Qa (orifice) flow rate ^b (m ³ /min)	Qa (orifice) [Ta] ^{1/2}
<u>NONE</u>	<u>5.5</u>	<u>33.34</u>	<u>721.66</u>	<u>0.9558</u>	<u>1.17</u>	
<u>18</u>	<u>5.1</u>	<u>42.86</u>	<u>712.14</u>	<u>0.9432</u>	<u>1.13</u>	
<u>13</u>	<u>4.9</u>	<u>49.40</u>	<u>705.60</u>	<u>0.9346</u>	<u>1.10</u>	
<u>10</u>	<u>4.5</u>	<u>60.79</u>	<u>694.20</u>	<u>0.9195</u>	<u>1.06</u>	
Operational Flow Rate		<u>26.20</u>	<u>728.80</u>	<u>0.953</u>	<u>—</u>	

^amm Hg = 25.4 (in. H₂O/13.6)

^b Qa (orifice) = $1/m \{[(\Delta H_2O) (Ta/Pa)]^{1/2} - b\}$

^c% Difference = $\left[\frac{Qa \text{ (sampler)} - Qa \text{ (orifice)}}{Qa \text{ (orifice)}} \right] [100]$

Sampler Calibration Relationship

☒ Lookup Table Validated (i.e., % difference ≤ 4)

☐ New calibration relationship:

$(X = \frac{Qa \text{ (orifice)}}{[Ta]^{1/2}}, Y = (P1/Pa))$

$m =$ _____ $b =$ _____ $r =$ _____

For subsequent calculation of sampler flow rate:

$Qa = \{[P1/Pa - b][Ta]^{1/2}\} \{1/m\}$

Operational Flow Rate 1.181 m³/min

Qa (Orifice)	Qa (sampler) (Lookup Table)	% Difference ^c
<u>1.17</u>	<u>1.154</u>	<u>-1.4</u>
<u>1.13</u>	<u>1.115</u>	<u>-1.3</u>

Figure 2.5. Example VFC sampler calibration data sheet.

- Ambient barometric pressure (Pa), mm Hg (or kPa).
- Ambient temperature (Ta), °C and K ($K = ^\circ C + 273$).
- Orifice S/N and calibration relationship.

Note: Consistency of temperature and barometric pressure units is required. It is recommended that all temperatures be expressed in kelvin ($K = ^\circ C + 273$). It is also recommended that all barometric pressures be expressed in either mm Hg or kPa (but do not mix the two units). Take care to avoid calibrating a PM10 sampler using one set of units and then performing sample calculations using another set of units.

7. Turn on the sampler and allow it to warm up to operating temperature (3 to 5 min).

Note: The sampler inlet may be partially lowered over the orifice transfer standard to act as a draft shield (if a shield is not otherwise provided). Use a block to provide at least 2 in. of clearance at the bottom for air flow and for the manometer tubing).

Then, read and record the orifice transfer standard's manometer reading, ΔH_2O , and the corresponding sampler relative stagnation pressure manometer reading, ΔP_{stg} , on the data sheet. (Relative stagnation pressure is a negative pressure [i.e., a vacuum] relative to atmospheric pressure as measured by a manometer with one leg open to the atmosphere.)

Note: Be sure to convert ΔP_{stg} to mm Hg using Equation 12 before recording the reading on the calibration data sheet:

$$\text{mm Hg} = 25.4(\text{in. H}_2\text{O}/13.6) \quad (\text{Eq. 12})$$

8. Install the other resistance plates or adjust the variable orifice value to obtain each of the other calibration flow rates and repeat Step 7 for each. At least four calibration flow rates are required, with at least three in the acceptable flow-rate range (i.e., 1.02 to 1.24 m³/min).

Difficulties may be encountered in obtaining flow rates in the acceptable range. Even with modified resistance plates (or with no plates) installed, it may be impossible to obtain three acceptable flow rates with a filter mounted on the sampler. In this case, either lower flow-rate calibration points must be used and the calibration must be extrapolated into the acceptable range, or additional calibration points must be run without a filter installed in the sampler. If additional calibration points are obtained without a filter, they should be examined carefully to make sure they are consistent with the calibration points obtained with a filter (i.e., they fall on a smooth curve through all the calibration points).

9. Plot the calibration data on a sheet of graph paper as specified in Step 5 of the next subsection.

Note: The data should be plotted in the field as the calibration is occurring, rather than afterwards back at the laboratory.

Repeat Step 7 for any data that are questionable on the plot. Running additional calibration points at differing flow rates or repeating the calibration points at the same flow rates is encouraged to improve the precision of the calibration.

10. Turn off the sampler and remove the orifice transfer standard.
11. Install a clean filter on the sampler in the normal sampling mode (use a filter cassette if one is normally used). Turn on the sampler and allow it to warm up to operating temperature.
12. Read the relative stagnation pressure as in Step 7 and record it on the data sheet in the row for the operational flow rate.
13. Perform the calibration calculations presented in the following subsection.

2.4.3 Calibration Calculations -

Gather together all the calibration data, including the orifice transfer standard's calibration information and the sampler calibration data sheet.

Note: These calculations should be done at the time of the calibration, rather than later. This approach will allow additional calibration points to be taken if questions arise about the data that have already been obtained.

1. Verify that the orifice transfer standard calibration relationship is current and traceable to an acceptable primary standard.
2. Calculate and record $Q_a(\text{orifice})$ for each calibration point from the orifice calibration information and Equation 6:

$$Q_a(\text{orifice}) = \{[\Delta H_2O(T_a/P_a)]^{1/2} - b\} \{1/m\} \quad (\text{Eq. 6})$$

where:

$Q_a(\text{orifice})$ = actual volumetric flow rate as indicated by the transfer standard orifice, m^3/min

ΔH_2O = pressure drop across the orifice, mm (or in.) H_2O

T_a = ambient temperature during use, K ($K = ^\circ\text{C} + 273$)

P_a = ambient barometric pressure during use, mm Hg (or kPa)

b = intercept of the orifice transfer standard's calibration relationship

m = slope of the orifice transfer standard's calibration relationship.

3. Calculate and record the value of the absolute stagnation pressure, P_1 , for each calibration point:

$$P_1 = P_a - \Delta P_{stg} \quad (\text{Eq. 13})$$

where:

P_1 = absolute stagnation pressure, mm Hg
 P_a = ambient barometric pressure, mm Hg
 ΔP_{stg} = relative stagnation pressure, mm Hg.

4. Calculate and record the stagnation pressure ratio:

$$\text{Stagnation pressure ratio} = P_1/P_a \quad (\text{Eq. 14})$$

5. On a sheet of graph paper, plot the calculated orifice transfer standard's flow rates, $Q_a(\text{orifice})$, on the x-axis vs. the corresponding stagnation pressure ratios, P_1/P_a , on the y-axis. Draw a smooth curve through the plotted data. If necessary, extrapolate the curve to include the acceptable flow-rate range.
6. If the sampler manufacturer has provided a factory calibration table (i.e., the lookup table) for the sampler, compare $Q_a(\text{orifice})$ for several points on the calibration plot with $Q_a(\text{sampler})$ determined from the factory calibration at T_a . Calculate the percentage difference between $Q_a(\text{orifice})$ and $Q_a(\text{sampler})$ using Equation 17:

$$\% \text{ Difference} = \left[\frac{Q_a(\text{sampler}) - Q_a(\text{orifice})}{Q_a(\text{orifice})} \right] [100] \quad (\text{Eq. 17})$$

If the agreement is within a few (i.e., 3 or 4) percent, the factory calibration is validated and may be used for subsequent sample periods. Proceed to Subsection 2.4.5.

7. If the agreement is not within a few percent, recheck the accuracy of the orifice transfer standard and recheck the calibration procedure. Look for leaks, manometer reading errors, incorrect temperature or pressure data, or miscalculations. Also check for abnormally low line voltage at the site (it should be at least 110 VAC), for the correct blower motor, and for the presence of a gasket between the motor and the choked-flow venturi. A factory calibration is not likely to be substantially incorrect, and any discrepancy of more than a few percent is probably due to some problem with the sampler or with the calibration procedure. However, if no errors or problems with the sampler or with the calibration can be found, or if no factory calibration is provided by the manufacturer, proceed as described in Subsection 2.4.4.

2.4.4 Generation of Calibration Relationship - VFC Sampler -

1. For each calibration point, calculate and record the quantity,

$$Q_a(\text{orifice})/[T_a]^{1/2}$$

where:

$Q_a(\text{orifice})$ = actual volumetric flow rate as indicated by the transfer standard orifice, m^3/min

T_a = ambient temperature during sampler calibration, K
($K = ^\circ\text{C} + 273$)

2. For the general linear regression model, $y = mx + b$, let $y = P_1/P_a$ and let $x = Q_a(\text{orifice})/[T_a]^{1/2}$, such that the model is given by:

$$P_1/P_a = m[Q_a(\text{orifice})]/[T_a]^{1/2} + b \quad (\text{Eq. 15})$$

Calculate the linear regression slope (m), intercept (b), and correlation coefficient (r).

Note: Inspect the plotted calibration curve to determine whether any of the calibration points that are substantially outside of the acceptable flow-rate range need to be eliminated so that they do not cause an inappropriate linear regression line.

3. For subsequent sample periods, the sampler's average actual operating flow rate, $\bar{Q}_a$, is calculated from the calibration slope and intercept using Equation 16:

$$\bar{Q}_a(\text{sampler}) = \{[(\bar{P}_1/\bar{P}_{av}) - b][T_{av}]^{1/2}\} \{1/m\} \quad (\text{Eq. 16})$$

where:

$\bar{Q}_a(\text{sampler})$ = the sampler's average actual flow rate, m^3/min

$\bar{P}_1/\bar{P}_{av}$ = average stagnation pressure ratio for the sampling period

T_{av} = average ambient temperature for the sampling period, K
($K = ^\circ\text{C} + 273$)

b = intercept of the sampler calibration relationship

m = slope of the sampler calibration relationship.

Note: The average value for P_1 should be calculated from stagnation pressure measurements taken before and after the sampling period. P_{av} should be estimated from barometric pressure for the sampling period. See also Subsection 3.3 for additional information.

4. If a calibration (lookup) table is desired, evaluate Equation 16 for various appropriate values of P_1/P_a and T_a and list the corresponding values of $\bar{Q}_a(\text{sampler})$ in tabular form.

Note: A calibration table based on Equation 16 may not match the manufacturer's lookup table exactly because the manufacturer may use a different or more sophisticated model than Equation 15.

2.4.5 Single-Point Operational Flow-Rate Verification -

This procedure compares the VFC sampler's normal operating flow rate to the design flow rate of the inlet (e.g., 1.13 m³/min).

1. Determine the value of P1/Pa for the operational flow rate obtained with only the filter cassette installed (Steps 11 and 12 of Subsection 2.4.2).
2. Determine the sampler's operational flow rate, Qa(sampler), that corresponds to this value of P1/Pa. Use the manufacturer's calibration table if it has been validated in Step 6 of Subsection 2.4.3; otherwise, use Equation 16.
3. Compare Qa(sampler) with the inlet design flow rate (e.g., 1.13 m³/min) using Equation 18:

$$\text{Design flow rate \% difference} = \left[\frac{Qa(\text{sampler}) - 1.13}{1.13} \right] [100] \quad (\text{Eq. 18})$$

This design flow rate percentage difference must be less than the allowable flow rate tolerance (i.e., ± 10 , if not otherwise specified by the manufacturer). However, this value should be well within ± 7 to allow for some variation with ambient temperature. If this value is not within ± 7 , recheck the calibration procedure and data for errors. Check the sampler for leaks, bad motor brushes, missing gaskets, incorrect motor type, or abnormally low line voltage. Because the VFC flow rate is not adjustable, the VFC manufacturer must be consulted to resolve cases of substantially incorrect VFC flow rates.

2.5 Sampler Calibration Frequency

To ensure accurate measurement of the PM10 concentrations, calibrate HV PM10 samplers upon installation and recalibrate as follows:

1. At least quarterly or annually according to the state's quality assurance program (see 40 CFR 58, Appendix A, for a description of the quality assurance requirements).
2. After any repairs that might affect sampler calibration (e.g., replacing the motor).
3. After relocation of the sampler to a different site.
4. If the results of a field flow-check exceed quality control limits (e.g., greater than ± 7 percent from the sampler's indicated flow rate).
5. Whenever a field flow-check or performance audit indicates that the sampler is out (or nearly out) of the acceptable flow-rate range (i.e., 1.02 to 1.24 m³/min) for the inlet.

Note: Multipoint flow-rate calibrations should be distinguished from single-point, quality control flow checks (see Subsection 3.5). The latter are done more frequently than calibrations and are intended to check if the sampler flow rate, $Q_a(\text{sampler})$, or the calibration relationship have changed significantly since the last calibration.

TABEL 2.1. EQUIPMENT CALIBRATION REQUIREMENTS

Equipment	Acceptance limits	Frequency and method of measurement	Action if requirements are not met	Calibration reference in Volume II, Section 2.2
Sampler	Indicated flow rate = true flow rate $\pm 4\%$.	Calibrate with certified transfer standard on receipt, after maintenance on sampler, and any time audits or flow checks deviate more than $\pm 7\%$ from the indicated flow rate or $\pm 10\%$ from the design flow rate.	Recalibrate.	None
Associated equipment				
Analytical balance	Indicated weight = true weight ± 0.0005 g.	Gravimetrically test-weigh at purchase and during periodic calibration checks; use three to five standard weights covering normal range of filter weights.	Have balance repaired and/or recalibrated.	Subsection 2.1
Relative humidity indicators ^a	Indicator reading = psychrometer reading $\pm 0\%$ RH.	Compare with reading of wet-bulb/dry-bulb psychrometer on receipt and at 6-mo intervals.	Adjust or replace to attain acceptance limits.	Subsection 2.2
On/off timer	± 30 min/24 h.	Check at purchase and routinely on sample-recovery days.	Adjust or repair.	Subsection 2.4
Elapsed-time meter ^b	± 2 min/24 h.	Compare with a standard timepiece of known accuracy at receipt and at 6-mo intervals.	Adjust or replace time indicator to attain acceptance limits.	Subsection 2.3
Flow-rate transfer standard (orifice device)	Indicated flow rate (from previous calibration) = true flow rate $\pm 2\%$.	Check at receipt and at 1-yr intervals against positive-displacement standard volume meter; recalibrate or replace orifice unit if damage is evident.	Adopt new calibration curve.	Subsection 2.5

^aThe relative humidity indicator is used in Subsection 4.3.

^bOptional with a continuous flow-rate recorder.